

# Criterion: Research Pipeline & Analytics Artifacts

## Architecture Decision Record

### Status

**Status:** Accepted

**Date:** 2026-01-05

### Context

The thesis requires dataset-based insights, but the product scope avoids real-time BI and heavy analytics infrastructure. The pipeline must be reproducible and reviewable.

### Decision

Keep research/ETL in Python under analysis/ with datasets committed for reproducibility. Generate plots/findings as static artifacts (SVG/PNG) intended to be embedded in the web UI.

### Alternatives Considered

| Alternative                | Pros                 | Cons                         | Why Not Chosen                   |
|----------------------------|----------------------|------------------------------|----------------------------------|
| Power BI/Fabric            | Rich dashboards      | Vendor + deployment overhead | Out of scope for MVP             |
| Frontend-only charts       | No separate pipeline | Hard reproducibility         | Analysis must be verifiable      |
| Warehouse + scheduled jobs | Scales well          | Too heavy                    | Not needed for read-only library |

### Consequences

**Positive:** transparent, in-repo workflow; low runtime complexity.

**Negative:** visuals are mostly static.

**Neutral:** pipeline can be formalized later (pinned env + standardized export).

## Implementation Details

### Key Implementation Decisions

- \* Separate research code from API runtime.
- \* Prefer deterministic exports over live analytics services.

### Project Structure

analysis/  
datasets/  
code\_solution/

## Requirements Checklist

| # | Requirement               | Status | Evidence/Notes                           |
|---|---------------------------|--------|------------------------------------------|
| 1 | Research code in repo     | [OK]   | analysis/code_solution/*                 |
| 2 | Datasets included         | [OK]   | analysis/datasets/*                      |
| 3 | Standard Python toolchain | [OK]   | pandas/numpy/matplotlib usage            |
| 4 | Exportable artifacts      | [!]    | Plots exist; export convention can be... |
| 5 | Not coupled to runtime    | [OK]   | No API dependency on Python pipeline     |

## Known Limitations

| Limitation               | Impact                  | Potential Solution                    |
|--------------------------|-------------------------|---------------------------------------|
| Dependencies not pinned  | Repro varies by machine | Add an analysis/requirements.txt      |
| No single outputs folder | Harder embedding        | Standardize analysis/outputs/ exports |